

ARUNKUMAR K R

✉ arun8778jul@gmail.com | ☎ +91 87787 65734 | 🌐 portfolio-blitz-eight.vercel.app | in arunkumar-kr
📞 arun-66102 | 📧 arunkumar_66 | 🏠 23cs020_kpriet

SUMMARY

AI Engineer and Backend Systems Developer specializing in end-to-end LLM application development, computer vision pipelines and production-grade API systems. Experienced in building multi-modal AI products using Python, FastAPI, LangChain, and cloud-hosted models. Targeting AI and ML roles where I can architect intelligent systems and the scalable infrastructure around them. Published researcher in deep learning-based industrial inspection (YOLOv8), RHCSA-certified, and competitive programmer with 500+ DSA problems solved.

EDUCATION

KPR Institute of Engineering and Technology , Coimbatore B.E. Computer Science and Engineering	Sept 2023 – Apr 2027 CGPA: 8.77 / 10
Kongu Vellalar Matriculation Higher Secondary School , Perundurai Higher Secondary Education (Class XII)	Jun 2022 – Mar 2023 94.6%

SKILLS

- **AI & ML:** LangChain, RAG, LLM Pipelines, CLIP, BLIP, BERT, TensorFlow, PyTorch, Hugging Face, LangGraph
- **Backend Development:** FastAPI, Node.js, Express.js, Django, REST APIs
- **Database & Infrastructure:** PostgreSQL, MongoDB, Redis, Linux, Git, Docker, Vector DBs
- **Languages & Frameworks:** Python, Java, JavaScript, C, SQL, React.js, LangChain, OpenCV
- **Soft Skills:** Problem Solving, Team Collaboration, Technical Communication, Leadership

EXPERIENCE

Reccsar Pvt Ltd <i>Backend Developer Intern</i>	Jan 2025 – Feb 2025
• Built and shipped 5+ production features using React.js, Node.js, and Express.js, improving operational workflows for 100+ active users. • Integrated 10+ REST APIs with a PostgreSQL backend, reducing manual data entry effort by ~30%.	
SmartED Innovations <i>Machine Learning Intern</i>	May 2025 – Aug 2025
• Built and trained supervised learning models (scikit-learn, TensorFlow/PyTorch) for classification & prediction task, achieving 95% accuracy/F1-score on a dataset of 1000 samples. • Designed an NLP pipeline for text classification, feature engineering and reducing manual review time by ~90%.	

PROJECTS

RouteX – AI-Powered Intelligent Document Processing & Routing <i>Python, FastAPI, React.js, PyMuPDF, Tesseract OCR, Groq API, Sentence Transformers</i>	Live Demo GitHub
• Developed a full-stack AI application that extracts text from digital and scanned PDFs using PyMuPDF with OCR fallback, generates concise summaries using Groq-hosted LLMs and classifies documents to the department. • Built a FastAPI REST backend with Server-Sent Events (SSE) progress streaming, automated email routing and a React + Vite frontend for real-time document processing and intelligent routing.	
ProManage – Scalable Project Management API <i>Node.js, TypeScript, PostgreSQL, Redis, JWT</i>	Live Demo GitHub
• Architected a REST API with Role-Based Access Control and JWT authentication, supporting distinct permission tiers for admins, managers, and members across concurrent teams. • Reduced API response time by 45% using Redis caching and Node.js Streams, sustaining low latency under 500+ concurrent requests.	

RESEARCH & ACHIEVEMENTS

- **Research Publication:** "Real-Time Surface Defect Detection in Metals using YOLOv8" – presented and published at the **3rd ICEAMST International Conference**; applied deep learning for real-time industrial defect classification.
- **RedHat Certified System Administrator (RHCSA)** – Recognized Linux system administration certification.
- **Building RAG apps using MongoDB** – Skill Badge by MongoDB University.
- **Competitive Programming:** Solved **500+ DSA problems** on LeetCode and HackerRank; certified in Problem Solving (Intermediate), Problem Solving (Basic), Java and SQL (Basic).
- **Leadership:** Executive Member, Dept. Association – organized technical events and workshops for 300+ students.